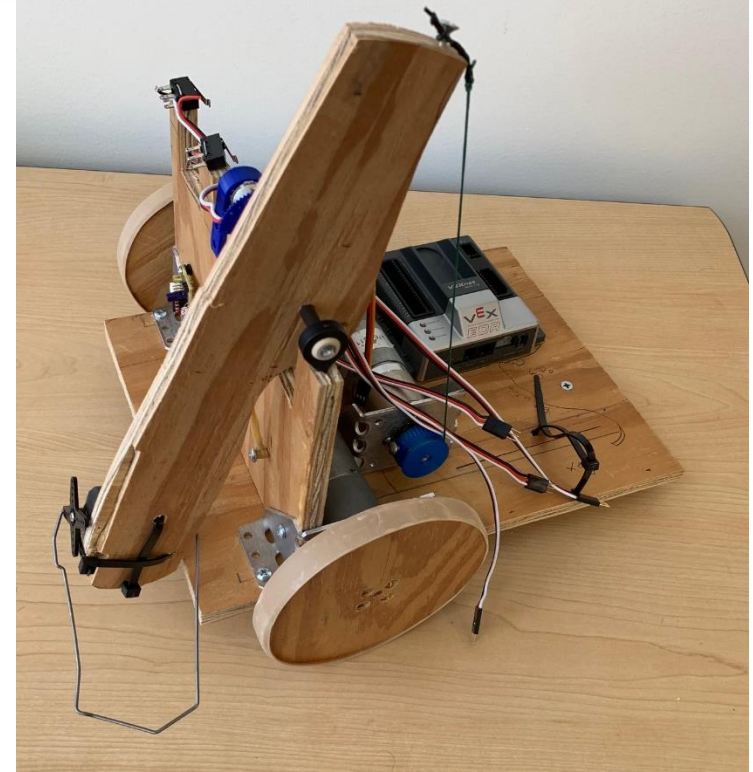


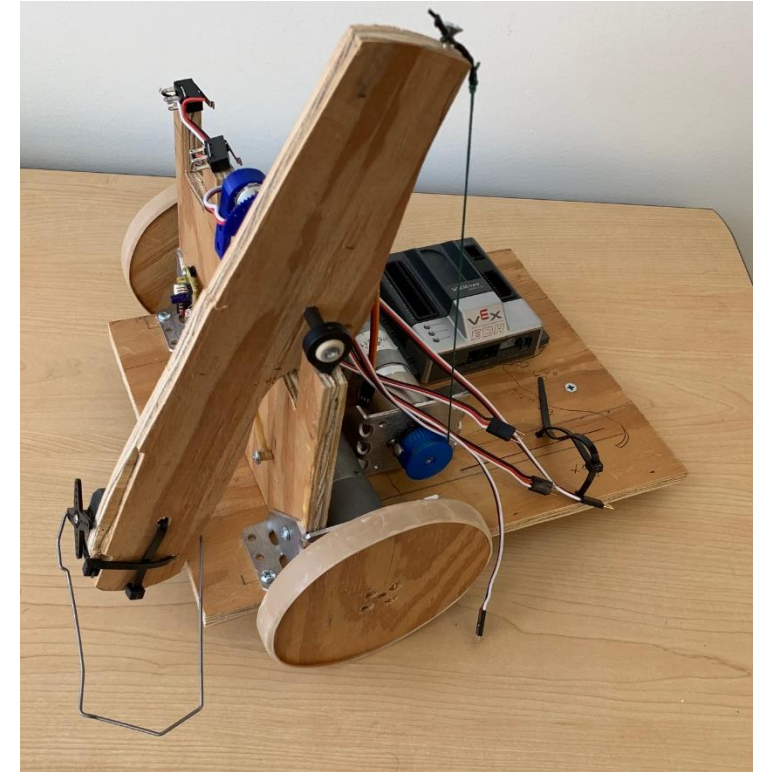
BEST SimBot Programming

By Student Competitions Team



Intro to SimBot

- Robot designed for teaching BEST workshops
- Built with BEST components
- Robot controller is a VEX Arm-Cortex Microcontroller
- Program using Simulink Support for VEX



SimBot Components

- A. 1 Servo Motor
- B. 3 DC Motors
 - Left Wheel
 - Right Wheel
 - Arm
- C. 2 Contact switches
- D. 2 Potentiometers
- E. 1 InfraRed (IR) Sensor

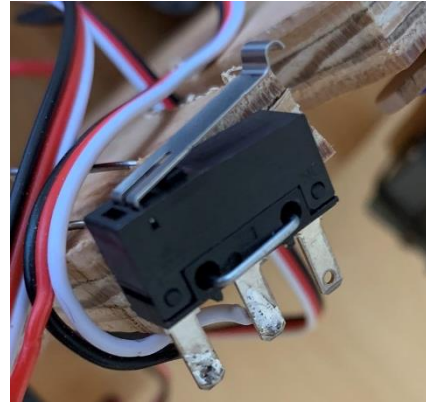
A



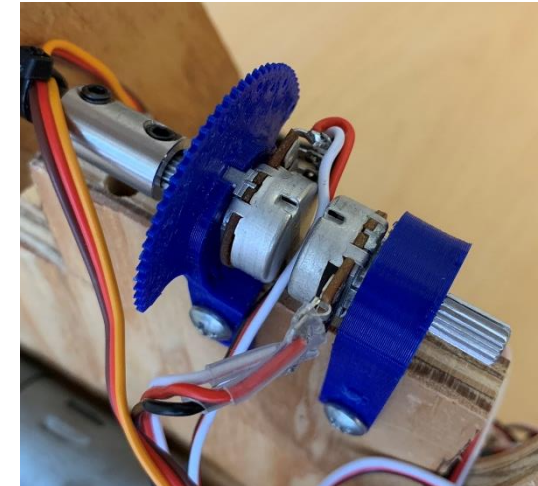
B



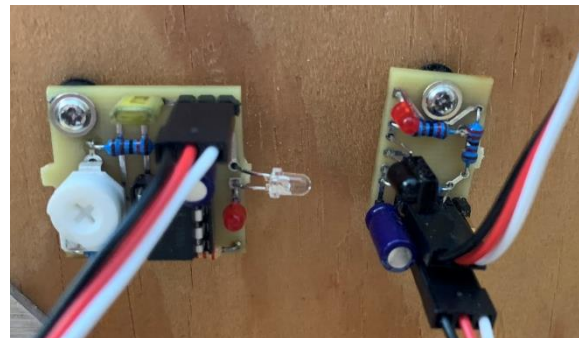
C



D



E



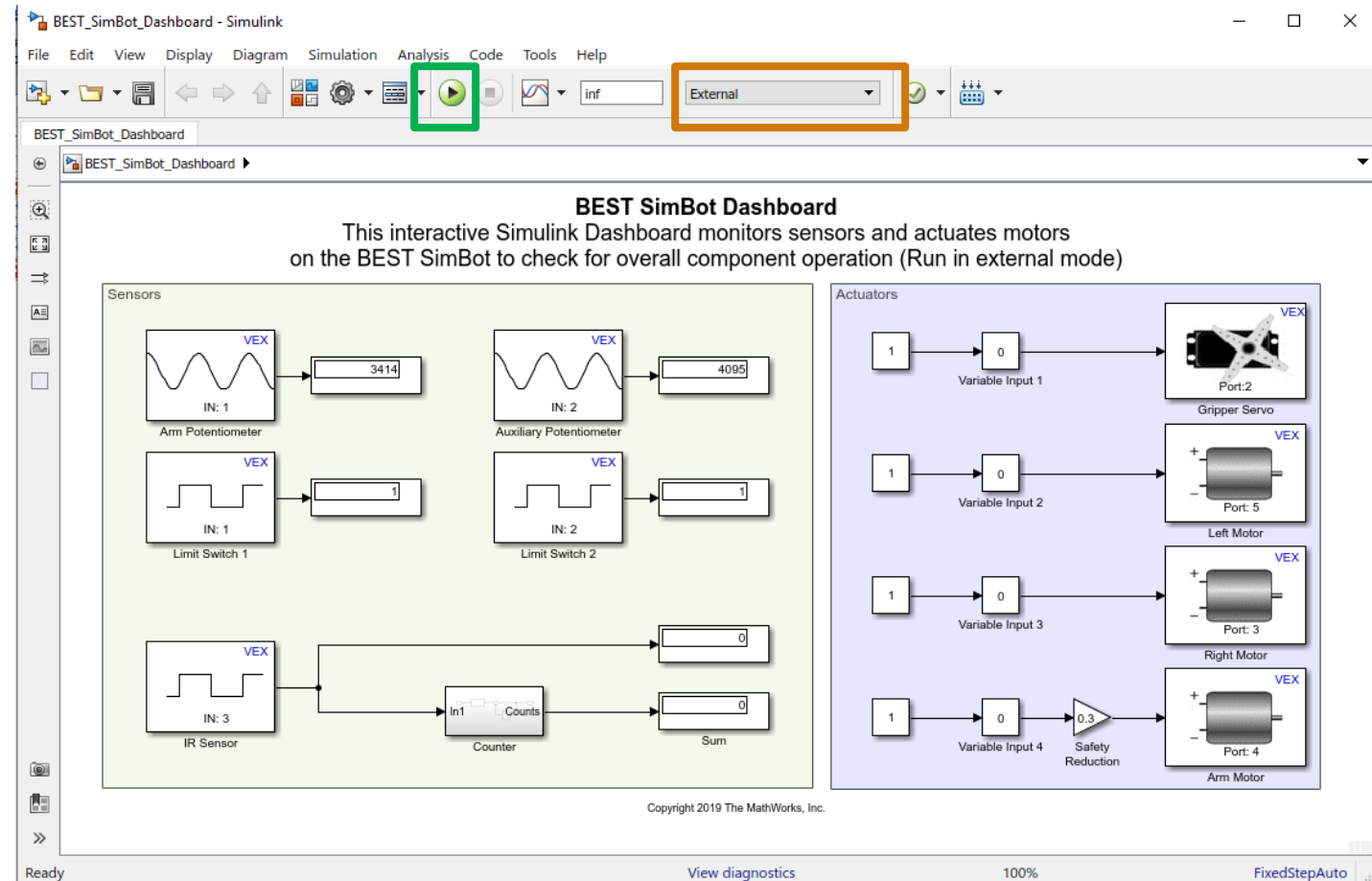
SimBot Lesson Plan

1. Interacting with Robot Sensors and Actuators
2. Controlling Robot Using Joystick
3. Programming Robot Autonomy

Interacting with Robot Sensors and Actuators

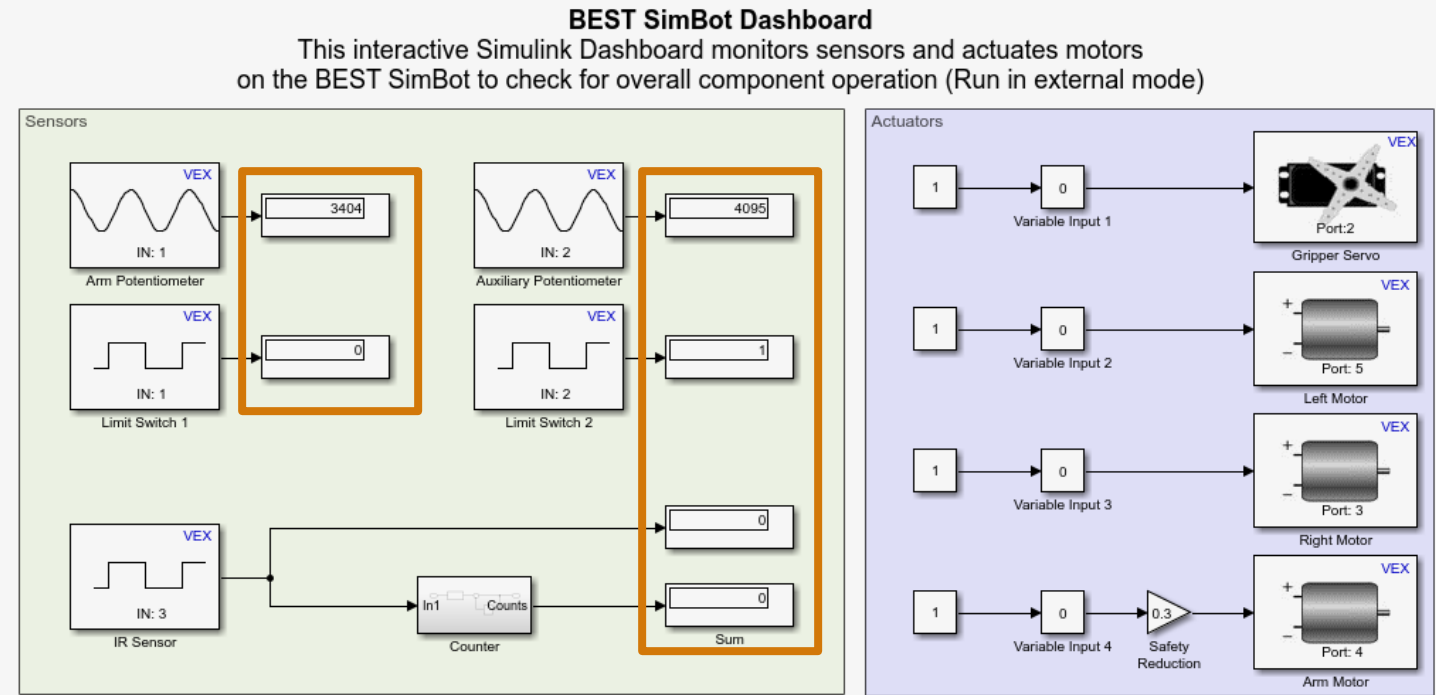
Running in External Mode

1. Open the Simulink model “BEST_SimBot_Dashboard.slx”
2. Make sure the VEX Cortex is connected
3. Run the model in external mode
4. Stop the model at any time using the “Stop” button

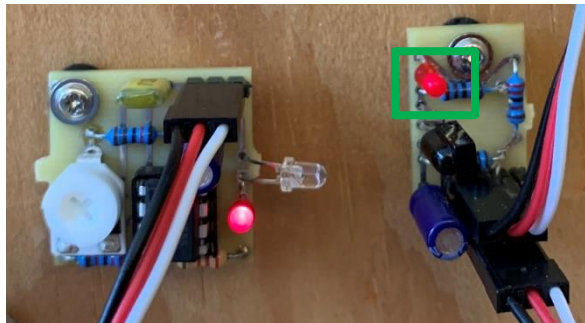


Check Sensor Functionality

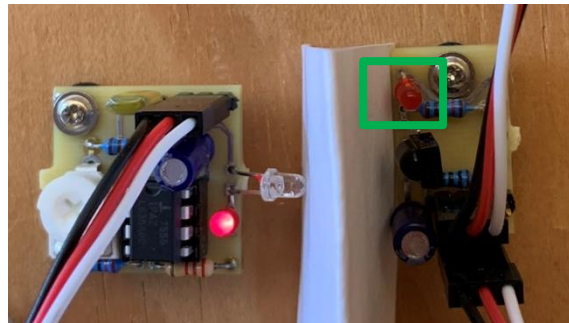
- Use the sensor displays to make sure the sensors are operating correctly
 1. Toggle contact switches
 2. Rotate potentiometers
 3. Break the beam on the IR sensor



IR Beam Active (LED ON)

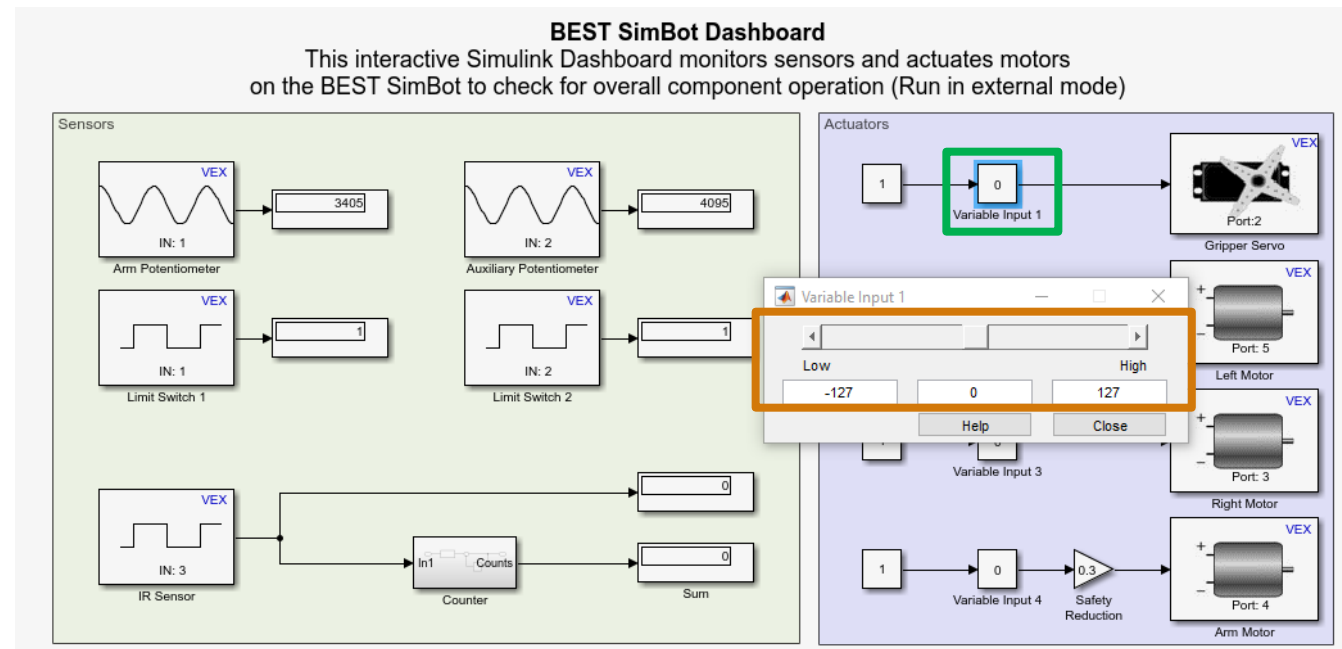


IR Beam Broken (LED OFF)



Writing Motor Commands

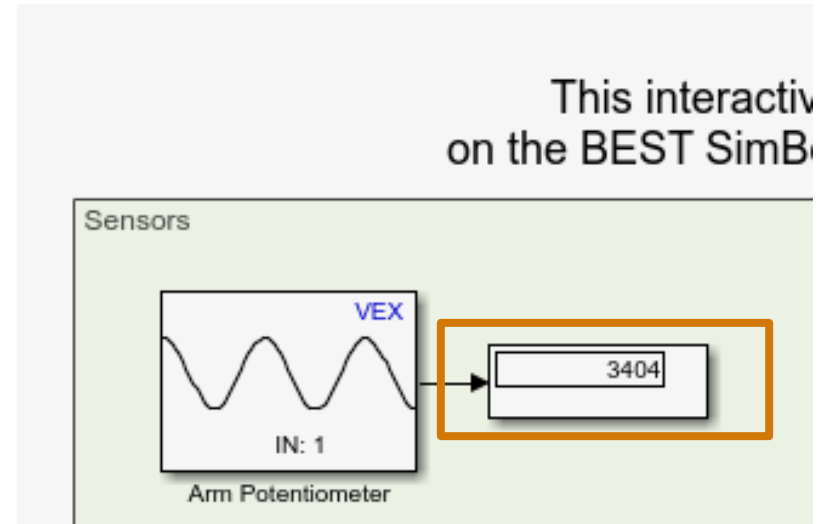
- Use the variable inputs to modify the commands sent to the motors
 1. Test Servo motor
 2. Test Wheel motors (Make sure robot is on stand or safe to move)
 3. Test Arm motor (Do this in small increments to be safe)



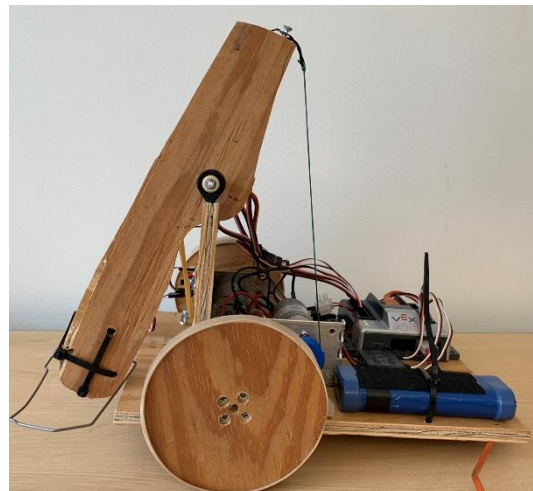
Record Max and Min Values for Arm

1. Use the display for the Arm Potentiometer to derive Minimum and Maximum arm positions that are safe to operate
2. The arm position should be calculated using the following equation.

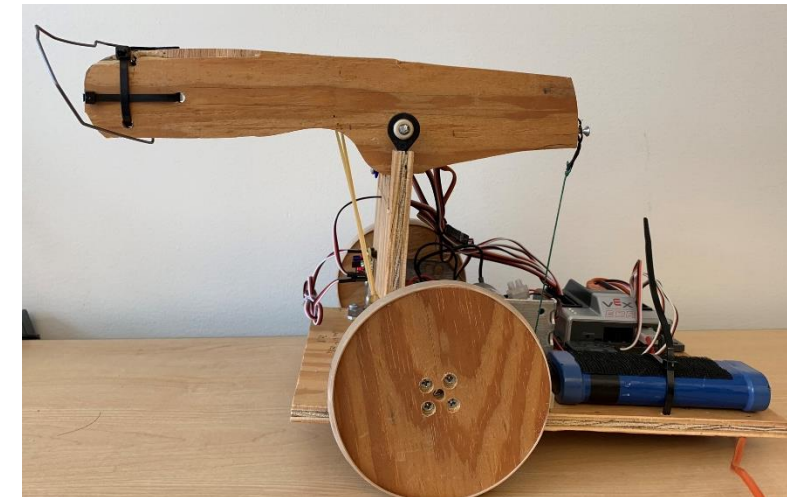
1. $Position = 4095 - Min/MaxValue$



Min Value

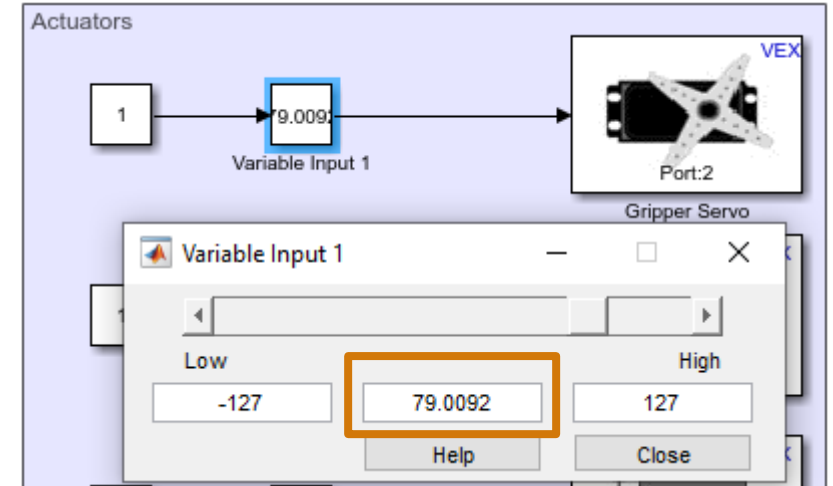


Max Value



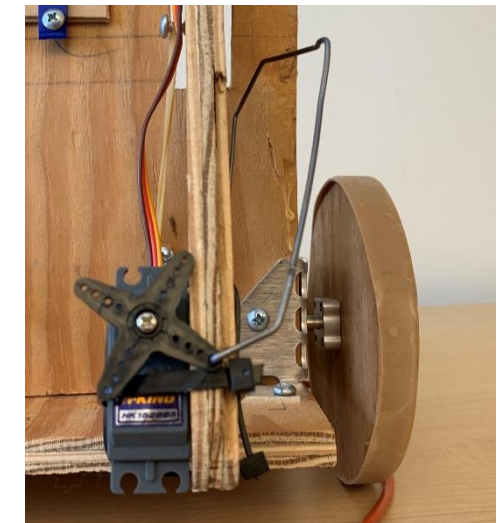
Record Max and Min Values for Servo

- Use the variable input for the Gripper Servo to derive Minimum and Maximum servo positions that are safe to operate



Max Value

Min Value



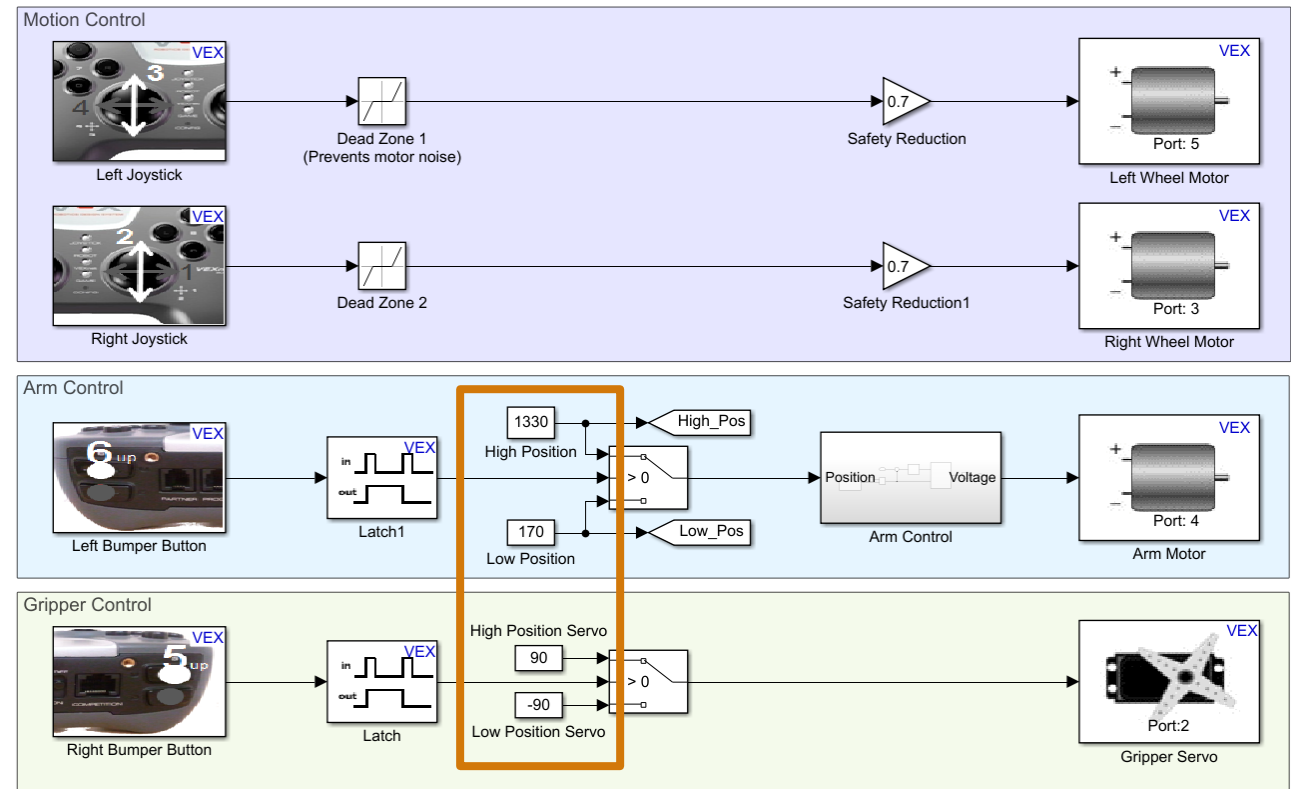
Controlling Robot Using Joystick

Control the Robot using the VEX Joystick

1. Open the Simulink model “BEST_SimBot_Joystick.slx”
2. Update the previously derived Min and Max motor positions for the Arm and Servo in the respective constant blocks

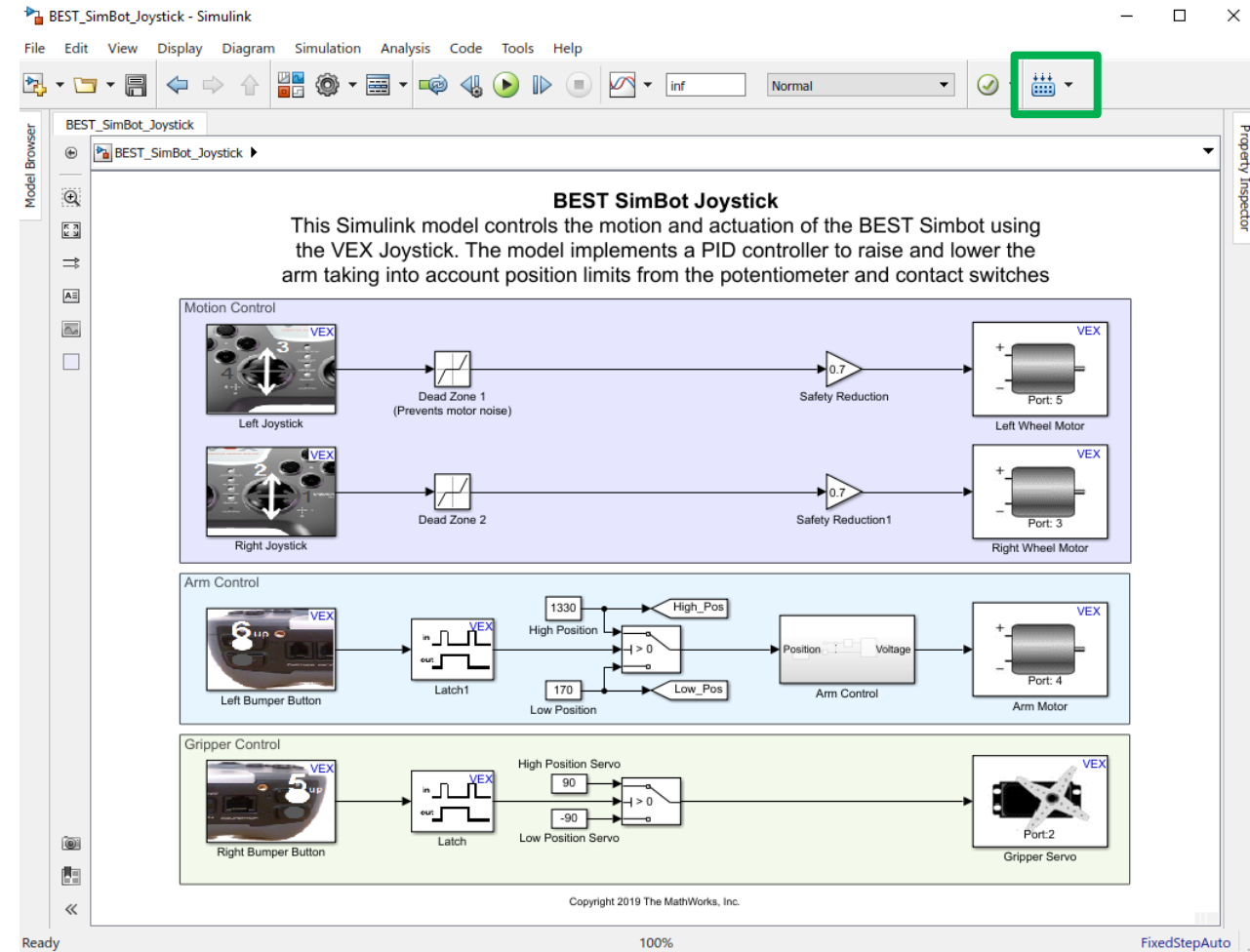
BEST SimBot Joystick

This Simulink model controls the motion and actuation of the BEST Simbot using the VEX Joystick. The model implements a PID controller to raise and lower the arm taking into account position limits from the potentiometer and contact switches



Deploy Model to VEX Microcontroller

1. Click the “Build” button in the model to upload the model to the robot
2. Connect the VEX Joystick to the Cortex
3. Verify functionality of all controls

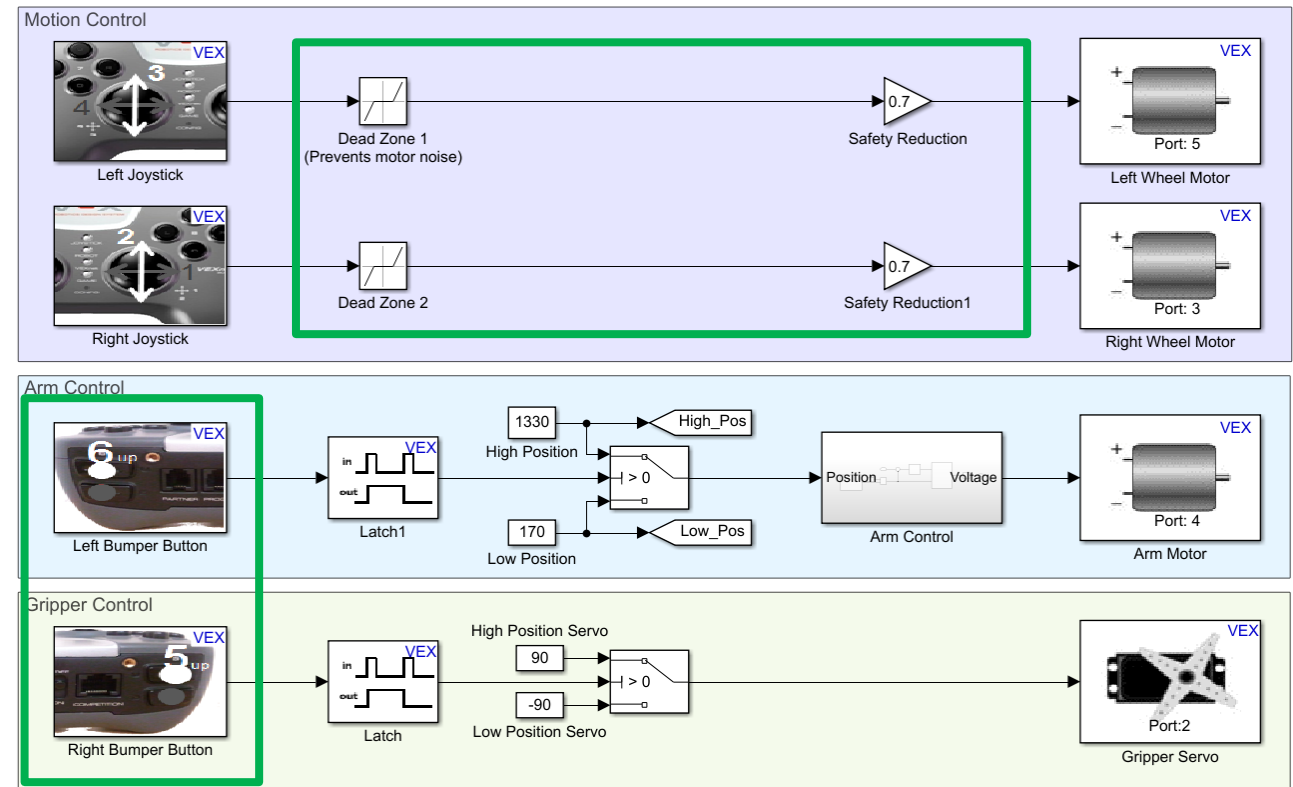


Exercise 1: Modify Model Parameters

- Modify the model by changing and testing the following
 1. Increase or decrease Safety Reduction gains
 2. Change the “Start” and “End” parameters of the Dead Zone blocks
 3. Select different buttons to control the Arm and Gripper

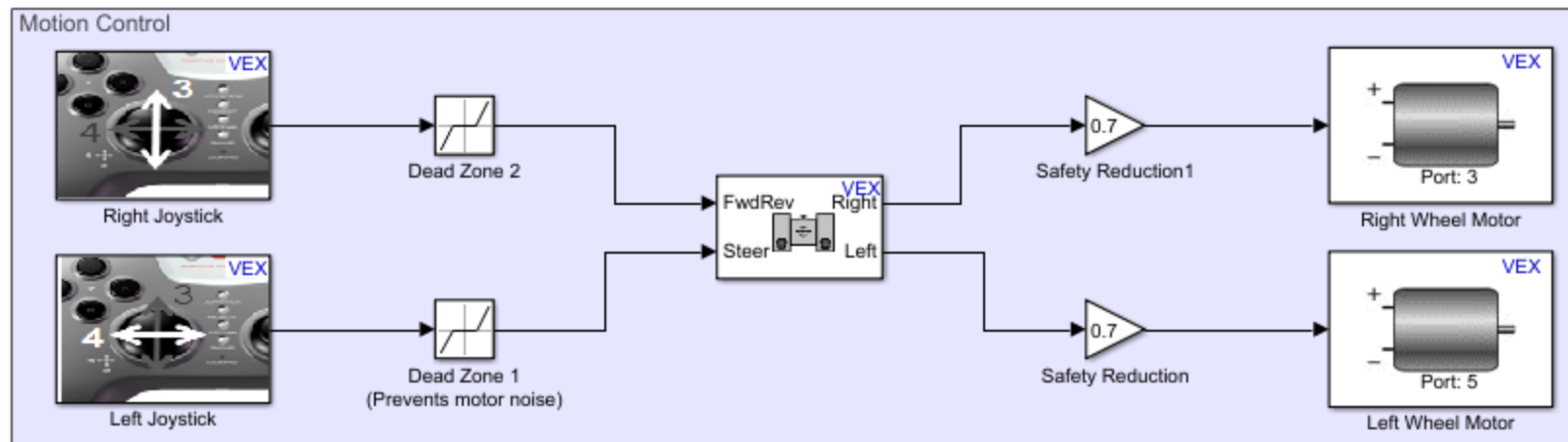
BEST SimBot Joystick

This Simulink model controls the motion and actuation of the BEST Simbot using the VEX Joystick. The model implements a PID controller to raise and lower the arm taking into account position limits from the potentiometer and contact switches



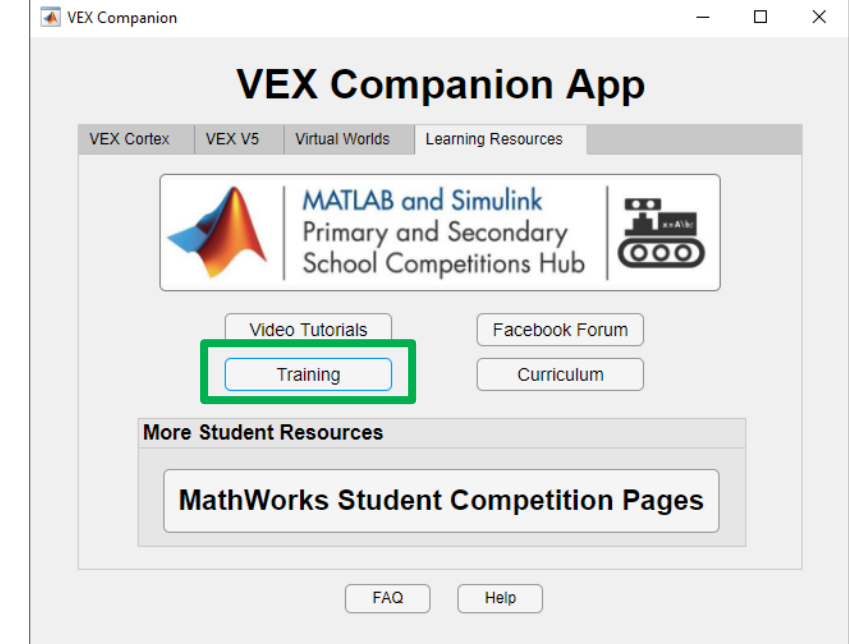
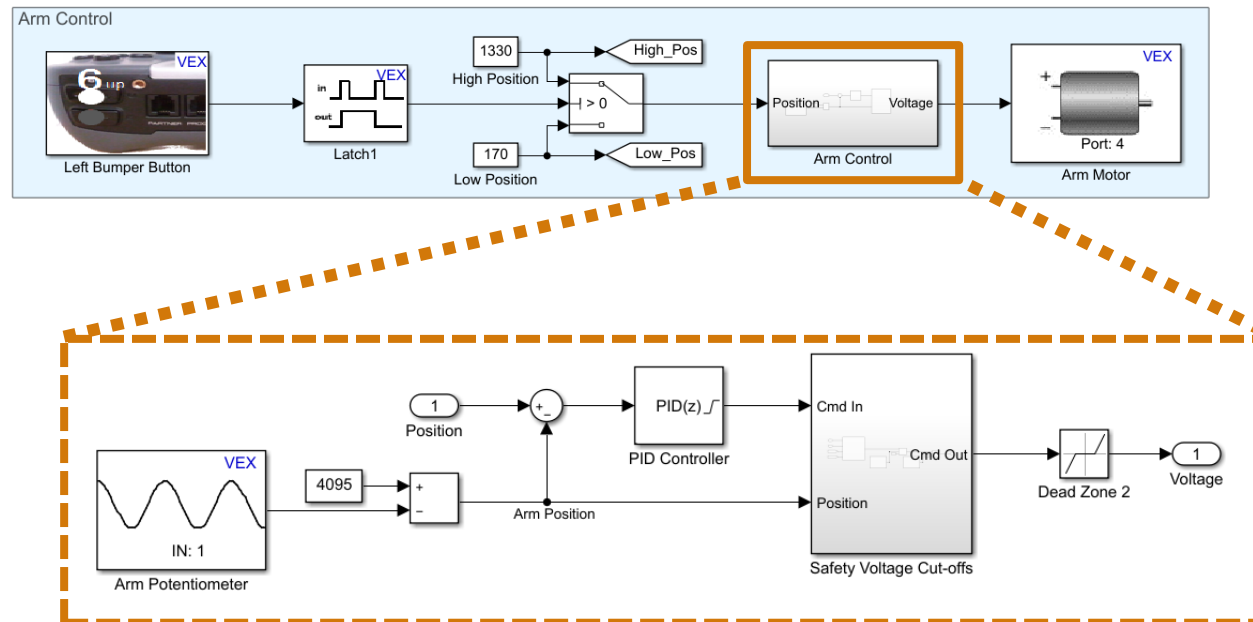
Exercise 2: Implement Arcade Control

1. Use the VEX Companion App to find the “Arcade Module” block
 - *VEX App>Open Cortex Library>Utilities>Arcade Module*
2. Connect the block to the joystick inputs and motor outputs
3. Change the joystick channels so only one joystick is necessary to control the robot motion



Arm Controller

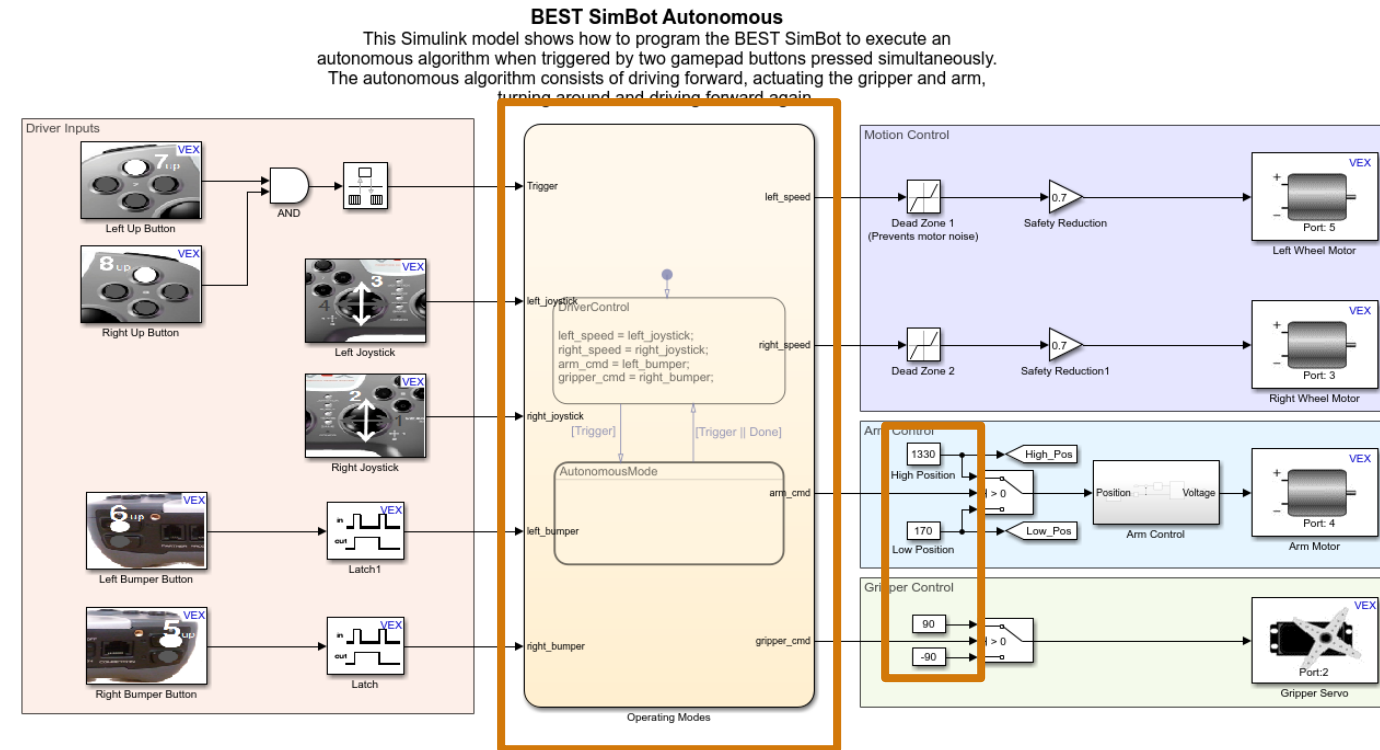
- The arm controller takes into account the information from the potentiometer connected to the arm to maintain a specified position.
- Potentiometer and contact switches are also used to cut-off the voltage at the arm limits.
- The controller implemented is a **PID algorithm**
- [Mobile Robotics Training](#) has lessons to help understand this algorithm



Programming Robot Autonomy

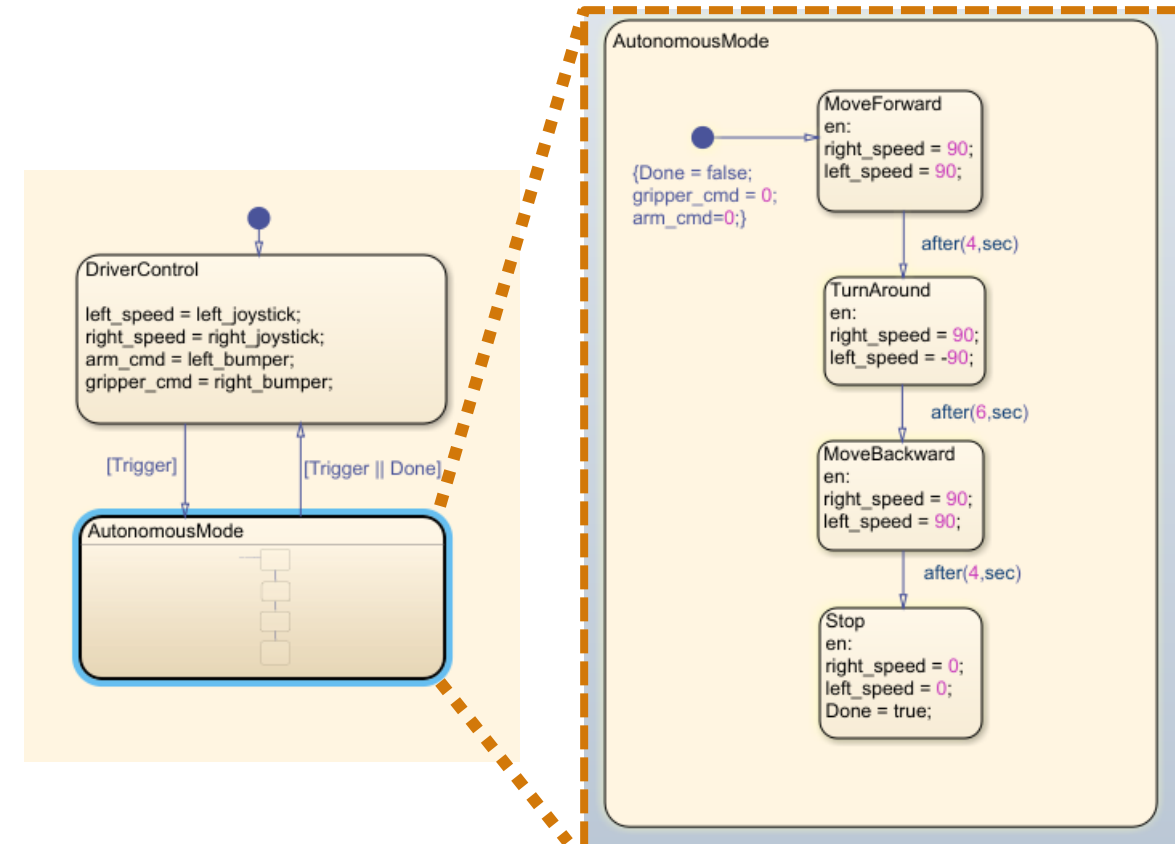
Updating the Autonomous Template

1. Open the Simulink model
“BEST_SimBot_Autonomous.slx”
 2. Update the Arm and Gripper Min and Max positions as in the previous section
- Note that all the actuator commands are now set through a Stateflow Chart



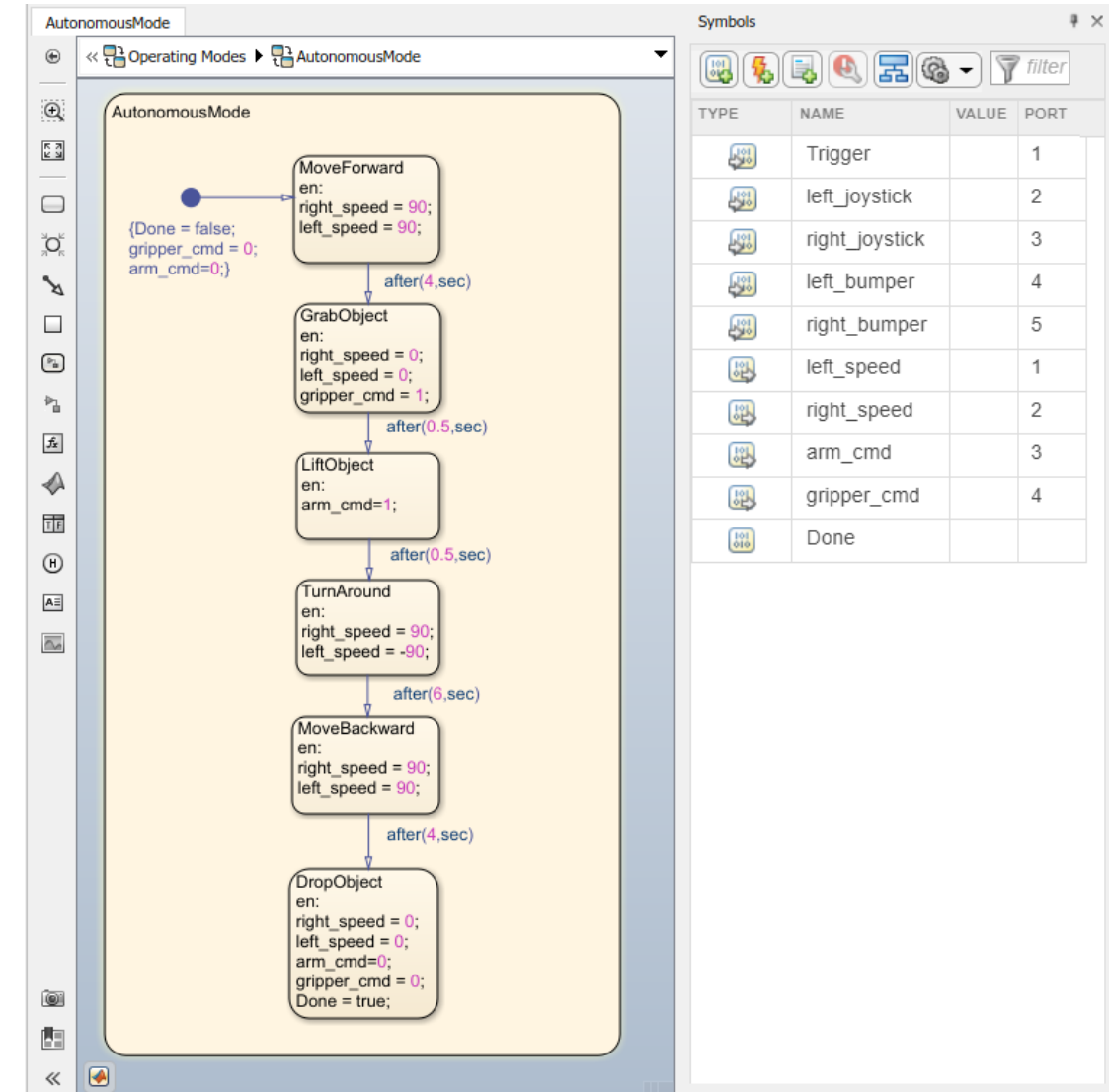
Upload the Autonomous Model

1. Click the “Build” button in Simulink to deploy the Autonomous template to the BEST SimBot
2. Navigate to the Autonomous Portion of the Stateflow chart to understand how the robot has been programmed to move forwards and backwards
3. Connect the VEX joystick to the Cortex
4. Verify that joystick control is still operational
5. Press the two buttons specified in the model as “Trigger” simultaneously to activate the autonomous mode



Exercise 3: Program Robot to Grab Object

- Modify the “Autonomous Mode” Stateflow Chart to have the robot automatically grab an object using the arm
- Create the necessary states and connections to account for the extra tasks
- Make sure that all the necessary variables are well defined as Symbols in the Stateflow chart.



End of SimBot Lessons!

- Congrats!

- Here are some learning outcomes from this lesson
 - How to use the VEX Support package for Simulink
 - How to navigate the VEX Companion App and VEX Libraries
 - How to read sensor values and command actuators using the VEX Cortex
 - How to control VEX Robots using the VEX Joystick
 - How use Stateflow Charts to Automate robot behavior